

SECTION B

7 (**short**) As shown in Fig. 6, a spinning ice skater is modelled using a central point mass, M (representing the body), connected by light links of length R to two point masses, each of mass m (representing the outstretched arms). The skater's initial angular velocity is Ω , and the frictional torque between the skates and the ice can be neglected.

(a) The skater pulls in her arms, after which the masses representing them are at a distance $R/2$ from the axis of spin. Find an expression for the new angular velocity of the skater. [5]

(b) If it takes time T for the skater to pull in her arms, find an expression for the average power required. [5]

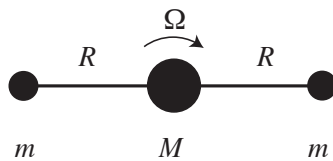


Fig. 6